

Scenario 1 — Search Functionality

✗ Weak: "Test the search feature."

What's missing: No product type (e-commerce, hospital), no filter/sort behavior, no empty/invalid input scenarios, no performance angle.

Response for Poor Prompt in CHATGPT

Test the search feature.

Response for Poor prompt in Qwen

The search feature is working perfectly!

To test it, I performed a sample web search for `***latest advancements in quantum computing 2026***`. The tool successfully retrieved recent and relevant results from across the web, including:

- Breakthroughs in quantum hardware, algorithms, and signs of "quantum advantage" [[1]].
- Research groups significantly reducing the qubits and time required to test online security technologies [[2]].
- Expanded pay-as-you-go access to quantum processing from major providers like AWS Braket, Google Quantum AI, IBM Quantum Network, and Microsoft Azure Quantum [[3]].
- IBM's 2026 quantum roadmap detailing new profiling tools and expectations for running circuits with up to three 120-qubit modules [[5]].

Let me know if you'd like to test a specific query, extract content from a particular URL, or run any other tests!

ICEPOT Framework

Context: The Environment is [amazon.com](https://www.amazon.com)

Instruction: Test the search feature for the Product "Shoe"

Constraint: Filter for women, size- 7 inch colour-green, Sort by Latest

Persona: As a Test Engineer

Output: CSV.

The Output shall be in below format.. Fetch the number of items, List down the number of items, Pass the test if above constraints are met and validate the performance

Tone: Professional, concise and data only.

Result in CHATGPT

Field	Result
Test ID	TC001
Environment	amazon.com
Product	Shoe
Gender Filter	Women
Size	7
Color	Green
Sort Order	Latest
Items Found	128
Items Listed	5
Constraints Met	No
Performance Validation	Search results were returned, but the available results did not verify all requested filters and Latest sorting on Amazon.com.
Test Result	FAIL

Result in QWENAI

****Test Execution Report****

- ****Environment****: amazon.com
- ****Product****: Shoe
- ****Applied Filters****: Gender: Women, Size: 7, Color: Green
- ****Sort Criteria****: Latest
- ****Total Items Found****: 10
- ****Items Listed****:
 1. Green - Women's Shoes (Sandals, Boots, Sneakers, Size 7 available)
 2. Paul Green - Women's Shoes (Hadley Sneakers, Size 7)
 3. Green Nike Shoes Women (Bella 7 Women's Workout Shoes, Size 7)
 4. Hunter Green Shoes For Women (Square Toe Ballet Flats, Size 7)
 5. adidas Campus 00s Mens Shoes Size 7, Color: Dark Green *(Constraint Violation: Men's)*
 6. YONEX Dark Green - Size 7 Fashion Sneakers
 7. Blue And Green Shoes (Nike Women's Pacific Shoes, Size 7)
 8. Hoka Bondi 9 Womens (Athletic shoes)
 9. Women's Flats - Grey - Size 7 *(Constraint Violation: Color Grey)*
 10. Women's Designer Shoes (General category listing)
- ****Constraints Met****: No
- ****Sort by Latest Verified****: No (External search API does not guarantee execution of Amazon's native "Sort by Latest: Date" parameter)
- ****Performance****: 142 ms (Search retrieval time)
- ****Test Status****: FAIL

****Defect Notes****:

1. Filter leakage detected (Men's product returned in women's filtered search).
2. Color filter leakage detected (Grey product returned in green filtered search).
3. Sorting parameter cannot be validated via standard web search endpoints; requires native UI automation.

Scenario 2 — File Upload

 Weak: "Generate test cases for file upload."

What's missing: No file types, no size limits,

no success/failure criteria, no security angle (malicious file upload).

Output for Week prompt in CHATGPT:

TC ID	Test Scenario	Test Data / Steps	Expected Result	Priority
TC-001	Upload a valid file	Upload a supported file within size limit	File uploads successfully	High
TC-002	Upload multiple valid files	Select multiple supported files	All files upload successfully	High
TC-003	Upload unsupported file type	Upload <code>.exe</code> / unsupported extension	Upload is rejected with appropriate error	High
TC-004	Upload file exceeding size limit	Upload file larger than configured limit	Upload is rejected with size-limit message	High
TC-005	Upload file at maximum size	Upload a file exactly at the allowed limit	File uploads successfully	High
TC-006	Upload empty file	Upload a 0-byte file	System accepts/rejects according to requirements	Medium
TC-007	Upload file with special characters	Filename: <code>test @#\$%.pdf</code>	File is handled correctly	Medium
TC-008	Upload file with very long filename	Use filename exceeding supported length	Appropriate validation/error is displayed	Medium
TC-009	Upload duplicate file	Upload the same file twice	System handles duplicate according to requirements	Medium
TC-010	Upload file with uppercase extension	<code>DOCUMENT.PDF</code>	File is accepted if extension is supported	Medium
TC-011	Upload file with misleading extension	Rename an unsupported file with a supported extension	File type is validated correctly; malicious/invalid file is rejected	High

TC-012	Cancel file selection	Select a file, then cancel the file picker	No file is uploaded	Medium
TC-013	Remove selected file	Select a file and click Remove/Delete	File is removed from selection	Medium
TC-014	Upload without selecting a file	Click Upload with no file selected	Appropriate validation message is shown	High
TC-015	Upload corrupted file	Upload a corrupted PDF/image/document	File is rejected or handled appropriately	High
TC-016	Upload during slow network	Start upload on a slow connection	Progress/status is displayed correctly; upload eventually succeeds or times out gracefully	Medium
TC-017	Network interruption	Disconnect network during upload	Upload fails gracefully with an appropriate message	High
TC-018	Refresh during upload	Refresh browser while upload is in progress	System handles the interrupted upload correctly	Medium
TC-019	Upload same file from different sessions	Upload identical file from separate sessions	System handles files independently according to requirements	Medium
TC-020	Verify uploaded file	Upload and then download/open the file	File contents remain unchanged	High
TC-021	Verify file name	Upload a valid file	Correct filename is displayed/stored	High
TC-022	Verify file size	Upload a file and check displayed size	Displayed size matches actual file size	Medium
TC-023	Verify upload progress	Upload a relatively large valid file	Progress indicator updates accurately	Medium
TC-024	Drag-and-drop upload	Drag a valid file into upload area	File is selected/uplo	

Output in Gemini:

Here is a comprehensive set of test cases for file upload functionality, organized by category for easy execution and tracking.

File Format & Type Validation

- **Valid File Formats:** Upload files with supported extensions (e.g., `.png`, `.jpg`, `.pdf`, `.docx`) and verify successful upload.

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- **Invalid File Formats:** Upload files with unsupported extensions (e.g., `.exe`, `.bat`, `.sh`, `.php`) and verify that an appropriate error message is displayed.
-
- **Multiple File Formats:** Upload a mix of valid and invalid files simultaneously (if batch upload is supported) and verify error handling.
-
- **MIME Type Spoofing:** Rename an executable file to a valid extension (e.g., changing `script.exe` to `image.png`) and verify the system detects the true file type via MIME checking rather than relying solely on the file extension.
-
- **Case Sensitivity:** Upload files with uppercase or mixed-case extensions (e.g., `.JPEG`, `.Pdf`) and verify they are processed correctly.
-
- **Missing Extension:** Upload a valid file with no extension and verify how the system handles it.
-

File Size Limits

- **Within Size Limit:** Upload a file well within the allowed size limit (e.g., a 1 MB file when the limit is 10 MB) and verify success.
-
- **Maximum Boundary Size:** Upload a file exactly at the maximum allowed size limit and verify success.
-
- **Exceeding Maximum Size:** Upload a file slightly above the size limit (e.g., 10.1 MB when the limit is 10 MB) and verify that the upload is rejected with a clear error message.
-
- **Zero-Byte File:** Upload an empty file (0 KB) and verify that the system rejects it or warns the user.
-
- **Large File Upload Interrupt:** Start uploading a large file and disconnect the network mid-way to verify proper error handling and cleanup.
-

File Name Validation

- **Special Characters in Name:** Upload files containing special characters (e.g., `file#name@!$%^&*().png`) and check if the name is sanitized or accepted safely.
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- **Spaces in Name:** Upload a file with spaces in its name (e.g., `my document test.pdf`) and verify it uploads and downloads without pathing issues.
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- **Very Long File Name:** Upload a file with an exceptionally long file name (e.g., 255+ characters) and verify the system handles or truncates it gracefully.
-
- **Duplicate File Names:** Upload a file with the same name as an already existing file and verify if it overwrites, renames (e.g., `file(1).png`), or prompts the user.
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- **Non-ASCII / Unicode Names:** Upload files named using non-Latin scripts (e.g., Chinese, Arabic, Cyrillic, accented characters) and verify correct display and retrieval.
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User Experience & UI

- **Drag-and-Drop:** Drag a file from the desktop directly into the drop zone and verify it triggers the upload.
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- **File Picker Navigation:** Click the upload button, select a file via the OS file explorer, and verify it attaches correctly.
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- **Progress Bar / Indicator:** Verify that a progress bar or spinner appears during upload, especially for larger files.
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- **Cancel Upload:** Start an upload and click the "Cancel" button before completion; verify the process stops and no partial file remains.
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- **Remove / Delete Attached File:** Upload a file, click the "Remove" or "Trash" icon before submitting, and verify the file is detached.
-
- **Multi-File Selection:** Attempt to select multiple files at once if multi-file upload is supported, or verify single-file restriction if only one is allowed.
-

Security Validation

- **Malicious Executable Files:** Attempt to upload scripts containing malicious payloads (e.g., `.php`, `.js`, `.html`, `.jsp`) to test for Remote Code Execution (RCE) or Cross-Site Scripting (XSS).
-
- **Path Traversal:** Name a file using relative path characters (e.g., `../etc/passwd` or `../../../../boot.ini`) to check if the server prevents writing outside the target directory.
-
- **Double Extensions:** Upload files with double extensions (e.g., `image.jpg.exe` or `document.pdf.php`) to test parser behavior.
-
- **Zip Bomb / Decompression Bomb:** Upload a small compressed file that expands into gigabytes of data to check if the server safely limits decompression size.
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- **File Overwrite Vulnerability:** Check if an unauthenticated user can overwrite files uploaded by other users by matching file paths or IDs.
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Database & Storage Integrity

- **Storage Verification:** Verify that the uploaded file is correctly saved in the designated directory or cloud bucket (e.g., AWS S3).
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- **Database Record:** Check if the file metadata (file name, size, upload timestamp, user ID, file path) is accurately recorded in the database.
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- **Download & Preview:** Download or view the uploaded file and verify that the file content matches the original without corruption.

ICEPOT FRamework

Content: Environment:Any Tamilnadu Government Websit

Success Criteria: When able to load successfully

Failure Criteria: Should not be able to upload any PII data without masking

Allowed file types: JPEG, PDF, .Doc only

Example: Indian Passport Seva Website

Instruction: Generate 5 test case with mandatory, optional, priority 1 and 2 test cases

Example, SNo, TestCaseName, Description, Test Steps, Expected Result, AC, US, Priority, Severity, Critical, Test Data, Pass/ Fail

Persona: Test Case Generator, Reviewer: Test Manager

Output: Show in tabular form and deterministic

Tone: Professional,Precise and Concise

Response from OpenAI:

Below are 5 deterministic test cases covering **mandatory/optional** scenarios and **Priority 1/Priority 2**, based on the stated file-upload requirements.

S N o	TestCa seNam e	Desc riptio n	Test Steps	Expect ed Result	AC	US	Pri orit y	Se ver ity	Cri tic al	Test Data	Pas s/ Fail
1	Upload valid JPEG without PII	Verify that a valid JPEG file witho ut PII can be uploa ded succe ssfull y. Mand atory	1. Open the Tamil Nadu Govern ment website. 2. Navigat e to the relevant upload form. 3. Select a valid JPEG	File is upload ed succes sfully and the user can procee d without an upload validati on error.	AC1: Valid JPE G is acce pted.	US1: As a user, I want to uploa d an accep ted file succe ssfull y.	P1	Hig h	Yes	docum ent.j pg — valid JPEG, no PII	Pas s/F ail

			file. 4. Click Upload/ Submit.								
2	Upload valid PDF without PII	Verify that a valid PDF file without PII can be uploaded successfully. Mandatory	1. Open the upload form. 2. Select a valid PDF file. 3. Click Upload/Submit.	PDF is uploaded successfully and the form proceeds to the next step.	AC2: Valid PDF is accepted.	US1	P1	High	Yes	document.pdf — valid PDF, no PII	Pass/Fail
3	Upload valid DOC without PII	Verify that a valid .doc file without PII can be uploaded successfully. Mandatory	1. Open the upload form. 2. Select a valid .doc file. 3. Click Upload/Submit.	.doc file is uploaded successfully.	AC3: Valid .doc file is accepted.	US1	P1	High	Yes	document.doc — valid DOC, no PII	Pass/Fail

4	Upload unmasked PII document	Verify that an uploaded document containing unmasked PII is rejected. Mandatory	1. Open the upload form. 2. Select a JPEG/PDF/DOC containing unmasked PII. 3. Click Upload/Submit.	System prevents the file from being uploaded and displays an appropriate validation/error message requiring PII to be masked.	AC4: Unmasked PII must not be uploaded.	US2: As a user, I must not be able to upload unmasked PII.	P1	Critical	Yes	Passport containing unmasked passport number, DOB, address, etc.	Pass/Fail
5	Upload optional file with unsupported type	Verify that an optional file uploaded rejects unsupported file types. Optional	1. Open the upload form. 2. Select an unsupported file such as .png or .xlsx . 3. Click Upload/Submit.	System rejects the file and displays an unsupported-file-type message. User can continue if the upload field is optional.	AC5: Only JPEG, PDF and .doc are accepted; optional field does not block submission when empty.	US3: As a user, I should be able to skip an optional upload.	P2	Medium	No	document.png — unsupported file type	Pass/Fail

Priority: P1 = highest-priority/core functionality; P2 = secondary functionality.

Reviewer: Test Manager

Persona: Test Case Generator / Reviewer

S N o	TestCaseName	Des cript ion	Test Step s	Ex pec ted Re sul t	A C	U S	Pr io rit y	Se ve rit y	Cr iti cal	T e s t D a t a	P a s s / F a il
1	TC_01_Valid_Mandatory_JPEG_Upload	Verify successful upload of a mandatory JPEG document without unmasked PII data.	1. Navigate to the Tamil Nadu Government portal.								

2. Go to the document upload section.
- 3.
3. Select a standard JPEG file containing no unmasked PII.
- 4.
4. Click Upload. | Page/Document loads successfully with a confirmation message. | AC_01 | US_01 | P1 | High | Yes | Masked_ID.jpeg (< 2MB) | Pass |

5. | 2 | TC_02_Valid_Optional_PDF_Upload | Verify successful upload of an optional PDF document without unmasked PII data. | 1. Access the upload section on the website.
- 6.
2. Locate the optional document field.
- 3.
3. Attach a valid PDF file with masked PII.
- 4.
4. Submit the file. | File attaches and loads successfully without errors. | AC_01 | US_01 | P2 | Medium | No | Sample_Doc.pdf | Pass |
5. | 3 | TC_03_Unmasked_PII_Passport_Failure | Verify failure criteria when attempting to upload a document containing unmasked PII (Passport). | 1. Open the document upload page.
- 6.
2. Select an unmasked Indian Passport image (JPEG format).
- 3.
3. Attempt to upload the file. | Upload is blocked; system displays error: "PII detected. Mask sensitive data before uploading." | AC_02 | US_01 | P1 | High | Yes | Unmasked_Passport.jpeg | Fail |
4. | 4 | TC_04_Invalid_FileType_Upload | Verify upload failure when an unsupported file type (e.g., .PNG) is selected. | 1. Navigate to the upload section.
- 5.
2. Select a file with an unallowed extension (.png).
- 3.
3. Click Upload. | System prevents selection or shows error: "Invalid file format. Allowed types: JPEG, PDF, .DOC." | AC_01 | US_01 | P2 | Medium | No | Document.png | Fail |
4. | 5 | TC_05_Valid_Doc_Upload_No_PII | Verify successful upload of a mandatory .DOC file containing no unmasked PII. | 1. Open the upload form.
- 5.
2. Browse and select a valid .DOC file with masked personal details.
- 3.
3. Trigger the upload action. | Document uploads successfully and displays in the application preview/status. | AC_01 | US_01 | P1 | High | Yes | Application_Form.doc | Pass |

Scenario 3— Shopping Cart

 Weak: "Test the cart functionality."

What's missing: No add/remove/update actions,
no price calculation validation, no coupon/discount scenarios, no out-of-stock handling.

Output from ChatGPT

TC ID	Test Scenario	Expected Result	Priority
TC-01	Add an in-stock product to cart	Product is added with correct name, price, and quantity	P1
TC-02	Add multiple products	All selected products appear correctly in the cart	P1
TC-03	Increase product quantity	Quantity and subtotal update correctly	P1
TC-04	Decrease product quantity	Quantity and subtotal update correctly	P1
TC-05	Remove a product	Product is removed and cart total is recalculated	P1
TC-06	Add the same product multiple times	Quantity is consolidated or handled according to requirements	P1
TC-07	Validate product price	Cart price matches the product/listing price	P1
TC-08	Validate subtotal	Subtotal = sum of item prices × quantities	P1
TC-09	Validate tax/shipping calculation	Applicable charges are calculated correctly	P1
TC-10	Validate final cart total	Final total is mathematically correct	P1

TC-1 1	Apply valid coupon	Discount is applied correctly	P1
TC-1 2	Apply invalid/expired coupon	Coupon is rejected with an appropriate message	P1
TC-1 3	Test minimum-value coupon	Discount is applied only when eligibility criteria are met	P2
TC-1 4	Add out-of-stock product	Product cannot be added, or appropriate stock message is displayed	P1
TC-1 5	Stock becomes unavailable after adding	Cart prevents checkout or informs the user appropriately	P1
TC-1 6	Set quantity to zero	Product is removed or validation is displayed	P2
TC-1 7	Enter quantity greater than available stock	System prevents invalid quantity and displays an appropriate message	P1
TC-1 8	Enter invalid/negative quantity	Invalid quantity is rejected	P2
TC-1 9	Refresh cart page	Cart contents remain consistent	P2
TC-2 0	Proceed to checkout with valid cart	User proceeds to checkout successfully	P1
TC-2 1	Checkout with empty cart	Checkout is prevented and appropriate message is displayed	P1
TC-2 2	Update/remove item during network/API failure	User receives an appropriate error and cart remains consistent	P2
TC-2 3	Verify cart after login/logout	Cart persistence follows the expected business rules	P2
TC-2 4	Verify price change after product is added	Updated price is handled according to business rules	P2

ICEPOT Framework:

Context- Test the shopping cart functionality

Instruction-

Consider below features

- Cart Actions
- Price Validation and Calculation

- Coupon/Discounts
- Out of Stock handling
- Quantity Validation
- Boundary Cases

Constraints-

Only Generate Priority 1 Mandatory Test Cases.

Tone- Keep it in Simple English and define in steps

Output Format-

SN, Test Case Description, Test Steps, Expected Result, Feature, Priority

Persona-

As a Senior Functional Tester from Consumer perspective

Below are the **Priority 1 mandatory test cases** for shopping-cart functionality, written from a consumer perspective with simple, step-by-step test steps.

S N	Test Case Description	Test Steps	Expected Result	Feature	Priorit y
1	Add an available product to the cart	1. Open the shopping website.2. Select an available product.3. Click Add to Cart .4. Open the cart.	The selected product is added with the correct name, price, and quantity.	Cart Actions	P1
2	Remove a product from the cart	1. Add a product to the cart.2. Open the cart.3. Click Remove .	The product is removed and the cart total is updated correctly.	Cart Actions	P1
3	Update product quantity	1. Add a product to the cart.2. Open the cart.3. Change the quantity.4. Save/update the cart.	The new quantity is displayed and the item subtotal and cart total are recalculated correctly.	Cart Actions	P1
4	Validate item price in cart	1. Note the product price on the product page.2. Add the product to the cart.3. Compare the product-page	The cart displays the correct product price.	Price Validation and Calculation	P1

		price with the cart price.			
5	Validate subtotal calculation	1. Add a product to the cart.2. Increase its quantity.3. Check the subtotal.	Subtotal equals unit price × quantity .	Price Validation and Calculation	P1
6	Validate total cart calculation	1. Add multiple products with different quantities.2. Open the cart.3. Check the total.	The total correctly reflects all item subtotals and applicable charges/discounts.	Price Validation and Calculation	P1
7	Apply a valid coupon	1. Add eligible products to the cart.2. Enter a valid coupon code.3. Click Apply .	The coupon is accepted and the correct discount is applied to the cart total.	Coupon/Discounts	P1
8	Reject an invalid coupon	1. Add a product to the cart.2. Enter an invalid coupon code.3. Click Apply .	The coupon is rejected and an appropriate error message is displayed. The total remains unchanged.	Coupon/Discounts	P1
9	Prevent adding an out-of-stock product	1. Open an out-of-stock product.2. Try to add it to the cart.	The product cannot be added, and an appropriate Out of Stock message is displayed.	Out of Stock Handling	P1
10	Handle product becoming out of stock	1. Add an available product to the cart.2. Make the product unavailable/out of stock.3. Return to or refresh the cart.	The cart clearly indicates that the product is unavailable and prevents an invalid checkout.	Out of Stock Handling	P1

11	Prevent quantity greater than available stock	1. Add a product to the cart.2. Increase the quantity beyond available stock.	The system prevents the quantity from exceeding available stock and displays an appropriate message.	Quantity Validation	P1
12	Validate minimum quantity	1. Add a product to the cart.2. Set the quantity to 0 or below the allowed minimum.	The system rejects the invalid quantity or removes the item according to the defined cart rules.	Quantity Validation	P1
13	Validate multiple products and quantities	1. Add multiple products.2. Set different quantities for each product.3. Review the cart.	Each product displays the correct quantity, subtotal, and overall cart total.	Boundary Cases	P1
14	Verify empty cart behavior	1. Add a product to the cart.2. Remove all products.3. Open/refresh the cart.	The cart shows an appropriate empty cart message and no incorrect total is displayed.	Boundary Cases	P1
15	Verify cart after page refresh	1. Add products to the cart.2. Refresh the cart page.	Cart items, quantities, prices, and total remain correct after refresh.	Boundary Cases	P1

Gemini Response

Prioritized Test Cases (Priority 1)

SN	Test Case Description	Test Steps	Expected Result	Feature	Priority

1	Add an available product to the cart	1. Open the shopping website. 2. Select an available product. 3. Click "Add to Cart". 4. Open the cart page.	The product is added to the cart with the correct name, price, and default quantity (1).	Cart Actions	P1
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2	Remove a product from the cart	1. Add a product to the cart. 2. Open the cart page. 3. Click the "Remove" button next to the product.	The product is removed from the cart, and the cart total updates to reflect the removal.	Cart Actions	P1
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3	Update product quantity in the cart	1. Add a product to the cart. 2. Open the cart page. 3. Increase the quantity field. 4. Click "Update Cart" (or verify auto-update).	The new quantity is displayed, and the item subtotal and total price recalculate correctly.	Cart Actions	P1
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4	Validate product unit price display	1. Note the price of a product on its product details page. 2. Add the product to the cart. 3. Open the cart page. 4. Compare the unit price in the cart with the product page price.	The unit price displayed in the cart matches the price listed on the product details page.	Price Validation and Calculation	P1
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5	Validate item subtotal calculation	1. Add a product to the cart. 2. Set the quantity to 3. 3. Verify the item subtotal.	The subtotal equals unit price multiplied by quantity ($\text{\text{Unit Price}} \times 3$).	Price Validation and Calculation	P1

6	Validate total cart calculation	1. Add two or more different products to the cart. 2. Open the cart page. 3. Verify the cart grand total.	The grand total accurately equals the sum of all individual item subtotals plus applicable standard taxes/fees.	Price Validation and Calculation	P1
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7	Apply a valid discount coupon	1. Add eligible products to the cart. 2. Enter a valid coupon code in the coupon field. 3. Click "Apply".	The coupon is accepted, the discount amount is shown, and the total price is reduced accordingly.	Coupon/Discounts	P1
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8	Reject an invalid discount coupon	1. Add a product to the cart. 2. Enter an invalid or expired coupon code. 3. Click "Apply".	An error message is displayed, the coupon is rejected, and the total price remains unchanged.	Coupon/Discounts	P1
9	Prevent adding an out-of-stock product	1. Navigate to an out-of-stock product page. 2. Attempt to click	The "Add to Cart" button is disabled or an "Out of Stock" warning appears, preventing addition.	Out of Stock handling	P1

		"Add to Cart".			
10	Handle item going out of stock while in cart	1. Add an available product to the cart. 2. Simulate the product going out of stock globally. 3. Refresh or view the cart page.	The cart highlights the item as unavailable and disables the checkout process until the item is removed.	Out of Stock handling	P1

11	Prevent increasing quantity beyond available stock	1. Add a product with limited stock (e.g., 5 items left) to the cart. 2. Attempt to set the quantity to a higher number (e.g., 6).	An error message appears stating stock limit is reached, and quantity resets to maximum available (5).	Quantity Validation	P1
12	Prevent setting quantity to zero or negative	1. Add a product to the cart. 2. Enter '0' or a negative number in the quantity field.	The system rejects the entry, displaying an error or automatically removing the item if set to 0.	Quantity Validation	P1

		3. Save or update the cart.			
13	Verify empty cart state	1. Ensure no items are in the cart (or remove all items). 2. Open the cart page.	An "Your cart is empty" message is displayed, and no invalid totals or broken UI elements appear.	Boundary Cases	P1
14	Verify cart state retention on page refresh	1. Add items to the cart. 2. Refresh the	All items, quantities, subtotals, and total remain identical after the page reloads.	Boundary Cases	P1

		browser page.			
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